

Shuaicheng Zhang

PHD CANDIDATE · COMPUTER SCIENCE DEPARTMENT OF VIRGINIA TECH

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Research Interest

My research centers on open-world graph learning — building graph learning models that can continuously discover and integrate new knowledge from multimodal data streams in real-time, enabling AI systems to evolve their understanding without retraining. This includes advancing graph-augmented knowledge representation and graph-based representation learning, with diverse applications in AI for finance, predictive maintenance, deep generative graph models, personalized federated learning, and relation extraction.

Education

Virginia Tech

PHD CANDIDATE COMPUTER SCIENCE

- Advisor: Dr. Dawei Zhou

Blacksburg, VA

August 2020 - present

MIT CSAIL

VISITING PHD STUDENT

- Advisor: Dr. Julian Shun

Cambridge, MA

May 2023 - Aug 2023

University of Maryland, College Park

MS TELECOMMUNICATIONS

- Advisor: Dr. Michael Dellomo

College Park, MD

August 2018 - May 2020

Virginia Tech

BS COMPUTER SCIENCE

- Minors in Mathematics

Blacksburg, VA

August 2013 - Dec 2017

Industrial Experience

- 2025 **Research Intern, Microsoft Research**, Redmond, WA, USA
- 2023 **Research Intern, MIT-IBM lab**, Cambridge, MA, USA
- 2021 **NLP Research Intern, Nature Language Processing Team**, Deloitte, NY, USA
- 2017 **Software Development Engineer Intern, Trust Team**, Hundsun Technologies Inc., Hangzhou, China

Preprints & Publications

Scientific Hypothesis Generation and Validation: Methods, Datasets, and Future Directions —Adithya Kulkarni, Fatimah Alotaibi, Xinyue Zeng, Longfeng Wu, Tong Zeng, Barry Menglong Yao, Minqian Liu, **Shuaicheng Zhang**, Lifu Huang, Dawei Zhou. In Arxiv, under review

CAPTAIN: Conformal-Prediction-Based Multi-Source Time-Series Forecasting —**Shuaicheng Zhang**, Tuo Wang, Stephen Adams, Sanmitra Bhattacharya, Sunil Reddy Tiyyagura, Edward Bowen, Balaji Veeramani, Dawei Zhou. In review of TMLR'2025.

PDMBench: A Standardized and Multi-Perspective Benchmark for Predictive Maintenance on Multi-Sensor Industrial Time-Series Data —**Shuaicheng Zhang**, Tuo Wang, Stephen Adams, Sanmitra Bhattacharya, Sunil Reddy Tiyyagura, Edward Bowen, Balaji Veeramani, Dawei Zhou. In review of ICLR'2025.

HeroFilter: Adaptive Spectral Graph Filter for Varying Heterophilic Relations —**Shuaicheng Zhang**, Haohui Wang, Junhong Ling, Xiaojie Guo, Yada Zhu, Si Zhang, Dongqi Fu, Dawei Zhou. (*Equal Contribution) In proceedings of (NeurIPS)'2025.

When Heterophily Meets Heterogeneity: New Graph Benchmarks and Effective Methods —Junhong Lin, Xiaojie Guo, **Shuaicheng Zhang**, Yada Zhu, Julian Shun. In Preceedings of the ACM **KDD**'2025. **Best Paper Award**

MentorPDM: Learning Data-Driven Curriculum for Multi-Modal Predictive Maintenance —**Shuaicheng Zhang**, Tuo Wang, Stephen Adams, Sanmitra Bhattacharya, Sunil Reddy Tiyyagura, Edward Bowen, Balaji Veeramani, Dawei Zhou. In Preceedings of the ACM **KDD**'2025.

UnifiedGT: Towards a Universal Framework of Transformers in Large-Scale Graph Learning —Junhong Lin, Xiaojie Guo, **Shuaicheng Zhang**, Yada Zhu, Dawei Zhou, Julian Shun. In Preceedings of the **IEEE BigData** 2025. **Patent Pending**.

TGEditor: Task-Guided Graph Editing for Augmenting Temporal Financial Transaction Networks —**Shuaicheng Zhang**, Yada Zhu, Dawei Zhou. In Proceedings of the ACM International Conference on AI in Finance(**ICAIF**)'2023(**oral**). **Patent Pending, Deployed at Wellsfargo**.

Personalized Federated Learning under Mixture of Distributions —**Shuaicheng Zhang***, Yue Wu*, Yanchi Liu, Quanquan Gu, Dawei Zhou, Haifeng Chen, Wei Cheng. (*Equal Contribution) In Proceedings of International Conference on Machine Learning(**ICML**)'2023.

Extracting Temporal Event Relation with Syntax-guided Graph Transformer —**Shuaicheng Zhang**, Ning Qiang, Lifu Huang. In Findings of North American Chapter of the Association for Computational Linguistics(**NAACL**)'2022.

Awards & Honors

- 2025 **KDD Best Paper Award in Benchmark and Dataset Track**, KDD 2025
- 2025 **KDD Outstanding Reviewer (Top 10%)**, KDD 2025
- 2024 **CCI SWVA Cyber Innovation Scholars**, Commonwealth Cyber Initiative (CCI) Southwest Virginia
- 2023 **CCI SWVA Cyber Innovation Scholars**, Commonwealth Cyber Initiative (CCI) Southwest Virginia
- 2022 **CIKM Travel Award**, CIKM 2022
- 2021 **Virginia Tech Pratt Fellowship**, Virginia Tech Department of Computer Science

Academic Service

CONFERENCE REVIEWER

- 2025 **AAAI 2025**, Program Committee
- 2025 **KDD 2025, ICML 2025, ICLR 2025, Neurips 2025**, Reviewer, Program Committee
- 2024 **KDD 2024, ICML 2024**, Reviewer, Program Committee
- 2023 **KDD 2023, ICDM 2023, WWW 2023**, Reviewer
- 2022 **SIGIR 2022, KDD 2022, NLPCC 2022, WWW 2023**, Reviewer
- 2021 **SDM 2022, WWW 2022, CIKM 2022**, Subreviewer
- 2021 **NLPCC 2021**, Reviewer

ORGANIZING COMMITTEE

- 2022 **The 1st Workshop on Trustworthy Learning on Graphs (TrustLOG) at The Conference on Information and Knowledge Management (CIKM)'2022**, Publicity Chair *Atlanta*

Skills

STATISTICS Advanced statistical analysis, inference, factor, regression analyses, hypothesis testing, hierarchical modeling

MACHINE LEARNING AND DEEP LEARNING TensorFlow, Keras, Pytorch, Scikit-Learn, XGBoost, Pandas, Numpy, Scipy, NLTK, NetworkX

PROGRAMMING Python, Java, C, Matlab, Shell, MySQL, HTML

BASICS Linux, Git, LaTeX, Docker

Teaching Experience

- Spring 22 **CS 5984 Computational Social Science**, Teaching Assistant
- Fall 21 **CS 5984 Natural Language Processing**, Teaching Assistant
- Spring 21 **CS 2505 Computer Architecture I**, Teaching Assistant
- Fall 20 **CS 3214 Operating System**, Teaching Assistant